

maclane 1

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March 2026

1 Axioms for Categories

Let $\mathcal{C} = (\mathcal{A}, \mathcal{B})$ be an arrows only metacategory. Sps u_1, u_2 are right units of f . Note $fu_1 = (fu_2)u_1 = f(u_2u_1)$, so $\langle u_2, u_1 \rangle \in \mathcal{B}$. Then $u_1 = u_2u_1 = u_2$, so the right unit is unique. Similarly, the left unit is unique. Denote by $\text{dom } f, \text{cod } f$ the right and left units, resp. Let $\mathcal{O} = \{\text{dom } f : f \in \mathcal{A}\} \cup \{\text{cod } f : f \in \mathcal{A}\}$.

I claim $\langle g, f \rangle \in \mathcal{B}$ iff $\text{cod } f = \text{dom } g$. Sps $\langle g, f \rangle \in \mathcal{B}$. Let $u = \text{dom } g$, so $\langle g, u \rangle \in \mathcal{B}$ and $gf = (gu)f = g(uf)$, so $\langle u, f \rangle \in \mathcal{B}$, so $u = \text{cod } f$ is the left unit of f . Conversely, sps $\text{cod } f = \text{dom } g = u$. Then $\langle g, u \rangle, \langle u, f \rangle \in \mathcal{B}$, so $\langle gu, f \rangle = \langle g, f \rangle \in \mathcal{B}$.

Let $T : C \rightarrow B, S : B \rightarrow A$ be full functors. Then $S \circ T : C \rightarrow A$ is a full functor.

Proof: Let $c, c' \in C$ and $h : STc \rightarrow STc'$ in A . Since S is a full functor, there exists $g : Tc \rightarrow Tc'$ in B such that $h = Sg$. Since T is a full functor, there exists $f : c \rightarrow c'$ such that $g = Tf$, so $h = STf$, so ST is full.

A functor $T : C \rightarrow B$ is faithful if $f_1, f_2 : c \rightarrow c'$ such that $Tf_1 = Tf_2$ implies $f_1 = f_2$. The forgetful functor $U : \text{Grp} \rightarrow \text{Set}$ is faithful, but not bijective nor full.

Sps FTSOC there is a functor $\text{Grp} \rightarrow \text{Ab}$ by $G \mapsto Z(G)$. Consider $S_2 \rightarrow S_3 \rightarrow S_2$. $S_2 \rightarrow S_3$ is inclusion and $S_3 \rightarrow S_2$ is the sign map with $(12) \mapsto (12)$. Then the composition $S_2 \rightarrow S_2$ is the identity of S_2 .

By functoriality, we have maps $\mathbb{Z}_2 \rightarrow \mathbb{Z}_3 \rightarrow \mathbb{Z}_2$. The maps $\mathbb{Z}_2 \rightarrow \mathbb{Z}_3, \mathbb{Z}_3 \rightarrow \mathbb{Z}_2$ are zero, so the composition $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ is zero. However, functoriality on the identity $S_2 \rightarrow S_2$ gives the identity map $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$, which is nonzero, absurd.

For each $G \in \text{Grp}$, pick $g \in G$ and let $\varphi_g \in \text{Inn } G$ be associated. For $f : G \rightarrow H$, define $T(f) = \varphi_h^{-1}f\varphi_g$. T is indeed a functor since $T(1_G) = \varphi_g^{-1}\varphi_g = 1_G$ and

for $f_1 : G_1 \rightarrow G_2$ and $f_2 : G_2 \rightarrow G_3$, we have

$$T(f_2 f_1) = \varphi_{g_3}^{-1} f_2 f_1 \varphi_{g_1} = \varphi_{g_3}^{-1} f_2 \varphi_{g_2} \varphi_{g_2}^{-1} f_1 \varphi_{g_1} = T(f_2) T(f_1),$$

so T is indeed a functor.

Take $G = S_3$, φ, f to be conjugation by $\sigma = (12), \tau = (23)$, resp, so we wish for $\varphi_{\sigma}^{-1} \varphi_{\tau} \varphi_{\sigma} \neq \varphi_{\tau}$, or $\varphi_{\sigma^{-1} \tau \sigma} = \varphi_{\tau}$, or $[\sigma, \tau] \notin Z(S_3) = 1$. We compute

$$[(12), (23)] = (12)(23)(12)(23) = (123)^2 = (132) \neq 1.$$

Claim: $\det_K$ is a natural transformation. Let $M_n(K)$ denote the set of $n \times n$ matrices over a commutative ring K . $M_n(K), K$ being monoids are categories over one object.